Electric and CNG vehicles

We are going to learn about Electric and CNG vehicles which is the future in transportation sector

Let’s begin with **What are Electric Vehicles?**

An Electric Vehicle is a vehicle that operates on an electric motor and uses electrical energy stored in batteries instead of an internal combustion engine that generates power by burning a mixture of fuel and gases.

Unlike vehicles with combustion engines electric vehicles do not produce exhaust gases during operation. This makes electric vehicles more environmentally friendly than the vehicles with conventional technology.

Electric vehicle is considered as a possible replacement for the current-generation automobiles in the near future to address environmental challenges.

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Let see how does **Electric Vehicles** work?

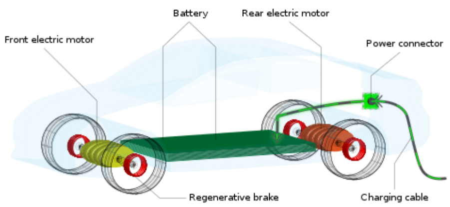
Electric Vehicles needs 3 important components

1. Controllers
2. Battery
3. Electric motor

When the pedal is pushed the controller gathers energy from the battery

Then Controller delivers the appropriate amount of electrical energy to the motor

Thus, this delivered electric energy transforms to mechanical energy therefore Wheels turn vehicles move.



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Electric Vehicles are not an new concept. In 1830 - first electric carriage was built.

In 1891 the first electric automobile was built in the USA.

Let’s discuss Types of electric vehicles
Four types of electric vehicles on the toad today

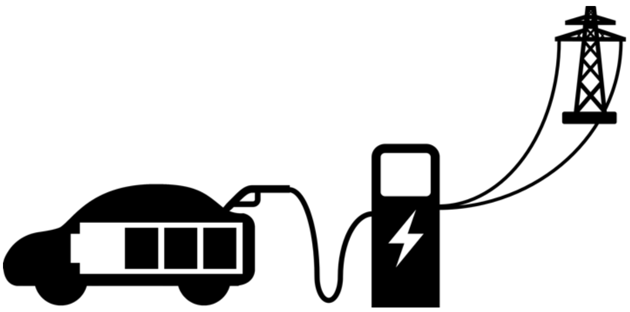
1. BEV that is Battery electric vehicles

2. PHEV that is Plug-in hybrid electric vehicle

3. HEV that is hybrid electric vehicle

4. Finally, FCEV that is Fuel-cell electric vehicle

Let’s discuss each one of them one by one
What is BEV: Battery electric vehicles.
Another name for Battery electric vehicles is All-Electric Vehicle (AEV)
A Battery electric vehicle runs entirely on a battery and without a need of an internal combustion engine. It is powered by electricity from an external source usually the public power grid. This electricity is stored in onboard batteries that turn the vehicles wheels using one or more electric motors.
BEVs can be charged at home overnight providing enough range for average journeys. However longer journeys or those who require a lot of hill climbs may require charging multiple times before you reach your destination.
The typical charging time for an electric car can range from 30 minutes and up to more than 12 hours. This all depends on the speed of the charging station and the size of the battery.
In the real world range is one of the biggest concerns for electric vehicles but is something that is being addressed by Research and Development and industry.



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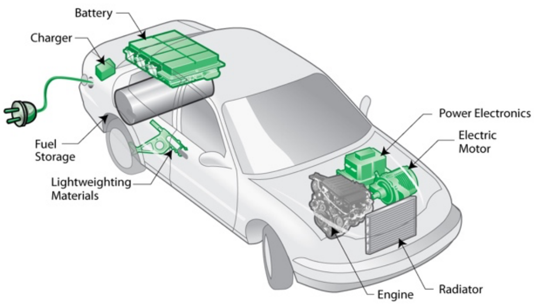
EV batteries are charged by plugging the vehicle into an electric power source.
EVs are far more efficient than conventional vehicles and produce no tailpipe emissions.
They also typically require less maintenance because the battery motor and associated electronics require little to no regular maintenance.
Further electric vehicles experience less brake wear thanks to regenerative braking systems.
Electric vehicles have fewer moving parts relative to conventional vehicles.
Electric vehicles do not contain the typical liquid fuel components such as a fuel pump fuel line or. fuel tank.

Next, we can discuss PHEV- Plug-in hybrid electric vehicle

Plug-in hybrid electric vehicle runs mostly on a battery that is recharged by plugging into the power grid. It is **also equipped** with an internal combustion engine which run on a gasoline or diesel fuel that can recharge the battery and/or to replace the electrical inverter when the battery is low and when more power is required
This makes them better for travelling long distances as you can switch to traditional fuels rather than having to find charge points to top up the battery.

PHEVs have smaller battery packs which means it can be used for medium range distances.
Of course, the same disadvantages that apply to combustion engine vehicles also apply to PHEVs such as the need for more maintenance engine noise emissions and the cost of petrol.



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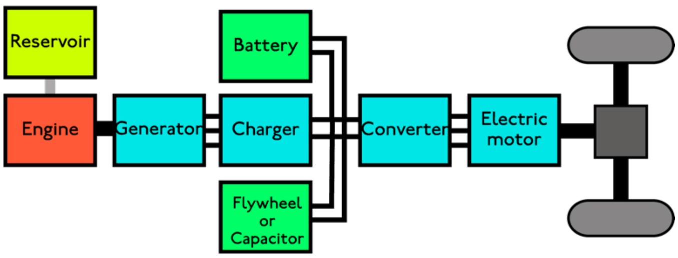
Next, we can discuss HEV Hybrid electric vehicle

An HEV has two complementary drive systems first one is a gasoline engine and fuel tank and the second one is an electric motor battery and controls. The engine and the motor can simultaneously turn the transmission which powers the wheels. The main difference between already discussed electric vehicles is HEV cannot be recharged from the power grid. Their energy comes entirely from gasoline and regenerative braking systems.



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Next, we can discuss FCEV - Fuel-cell electric vehicle more often it is called as hydrogen fuel cell electric vehicle.

A FCEV creates electricity from hydrogen and oxygen instead of storing and releasing energy like a rechargeable battery. Because of vehicles efficiency and water-only emission most of the experts consider these cars to be the best electric vehicles even though they are still in development phases.

Let’s discuss about the Advantages and disadvantages of Electric vehicles
Electric vehicles have low running costs as they have fewer moving parts for maintaining Electric vehicles and also environmentally friendly as they use little or no fossil fuels like petrol or diesel.
Compared to an internal combustion engine battery powered electric vehicles have approximately 99% fewer moving parts that need maintenance. Also, there is no need to lubricate the engines.
Electric vehicles create very little noise. Electric cars put a control on noise pollution as they are much quieter.
In electric vehicles there is no exhaust no spark plugs no clutch or gears.
Finally, it is Easy Driving – you can operate an electric car with just the accelerator pedal brake pedal and steering wheel.

Electric vehicles don’t burn fossil fuels instead uses rechargeable batteries.
Electric vehicles are energy efficient
For examples in electric vehicles batteries convert 59 to 62 percent of energy into vehicle movement while gas powered vehicles only convert between 17 and 21 percent.
Electric cars reduce emission. Electric cars are 100 percent eco-friendly as they run on electrically powered engines. Emission reduction including reduced usage of fuel is another advantage for all-electric vehicles. Because they rely on a rechargeable battery
Electric cars are high performance and low maintenance
The driving experience can also be fun because AEV motors react quickly making them responsive with good torque. AEVs are digitally connected with charging stations providing the option to control charging from an even a mobile app.
Finally electrical vehicles are Safe to Drive. - An electric car is safer to use given their lower center of gravity which makes them much more stable on the road in case of a collision.

Disadvantages of electric vehicles
The Initial Investment on **electric vehicles** is Expensive. Keep in mind Electricity isn’t Free.
Electric cars can travel only less distances. AEVs on average have a shorter range than gas-powered cars. Most models ranging between 100 and 200 km per charge and some luxury models reaching ranges of 300 miles per charge. This may be an issue **when looking at AEVs if you frequently take long trips.** Availability of charging stations can make AEVs less suitable for activities like road trips.
Electric cars take longer to “refuel”.
Fueling an all-electric car can also be an issue. Recharge Points or Electric fueling stations are still in the development stages.
Fully recharging the battery pack can take up to 8 hours and even fast charging stations take 30 minutes to charge. Thus, Electric car drivers have to plan more carefully because running out of power can’t be solved by a quick stop at the charging stations.

Electric cars are more expensive and battery packs may need to be replaced. Depending on the type and usage of battery batteries of almost all electric cars are required to be replaced every 3-10 years.

The battery packs within an electric car are expensive and may need to be replaced more than once over the lifetime of the car. These All-electric vehicles are more expensive than gas-powered cars.
Overall all-electric vehicles like any vehicle must be assessed based on personal needs and vehicle usage. There are many pros to owning an electric vehicle such as fuel savings and reduced emissions, but this can come at the cost of relying on battery charging and higher costs.
Electric vehicles create very little noise. Silence can be a bit disadvantage as people like to hear the noise if they are coming from behind them. Therefore, it can lead to accidents in some cases.
There are still challenges with electric vehicle batteries as they can experience thermal runaway which have for example caused fires or explosions.

Let's move to the second topic CNG operated vehicles
What is CNG?
CNG also known as compressed natural gas It is an eco-friendly alternative to gasoline. CNGs are made by compressing natural gas for example methane down to less than 1% of its volume at standard atmospheric pressure or compressed to 3000 - 3600 psi. CNG fuel is safer than gasoline and diesel because it is non-toxic. This natural gas is the same gas that you use daily to cook on the stove. The use of CNG fuel is becoming more popular with both commercial and non-commercial vehicles.
Compressed natural gas is under more pressure and thus takes up a smaller volume than ordinary natural gas.
Compressed Natural Gas is colorless non-carcinogenic and non-toxic. CNG is inflammable and lighter than air.
CNG is superior to petrol it operates at one-third the cost of conventional fuel and hence increasingly becoming popular with automobile owners. Commonly referred to as the green fuel because of its lead free characteristic and it reduces harmful emissions and is non-corrosive.
CNG is comprised of mostly methane gas. When CNG reaches the combustion chamber it mixes with air will be ignited by a spark and generates energy which moves the vehicle.
Please remember it is not a liquid fuel and is not the same as LPG Liquified Petroleum Gas which consists of propane and butane in liquid form. Also, CNG is not to be confused with liquefied natural gas.

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Why CNG is better than traditional petrol?
CNG is one of the most viable alternatives to traditional liquid fuels for vehicles.
CNG is one fifth the price of gasoline resulting in substantial savings in fuel costs.
CNG reduces maintenance costs since it contains no additives and burns cleanly leaving no by-products of combustion to contaminate your spark plugs and engine oil.
The engine oil also remains clean which minimizes engine wear and requires less frequent changes.
CNG is more environment friendly and CNG engines are much quieter due to the higher octane rating over gasoline.
CNG produces less exhaust emissions and as a result harmful emissions such as carbon monoxide carbon dioxide and nitrous oxide are generally reduced by as much as 95% when compared to gasoline powered vehicles.

Disadvantages of CNG:
CNG Gas stations have limited availability. In India some states have high number of CNG fuel stations.
CNG tank requires large space, and it is heavy. So, it affects reliability and vehicle performance.
Another issue with CNG vehicles is a longer breaking distance **due to the added weight of the fuel storage system.**
Further the composition of natural gas itself can be an issue. CNG is mainly comprises of methane which is a greenhouse gas which could contribute to climate change if a leak existed.